

CSE423: Computer Graphics

Assignment 3

Total Marks: 5

A digital artist is designing a vibrant banner with a gradient that transitions from **orange (RGB: 255, 165, 0)** to **pink (RGB: 255, 105, 180)**. The artist wants to manipulate the gradient using different color models (RGB, HSV, HLS) to control the hue, saturation, and lightness for a smooth and visually appealing transition.

1. **Convert RGB (255, 165, 0) to HSV and RGB (255, 105, 180) to HSV.**
 - Show all intermediate steps to calculate **Hue**, **Saturation**, and **Value** for both colors.
2. **Convert the resulting HSV values back to RGB for verification.**
 - Verify that the RGB values after conversion are consistent with the original input values.
3. **Convert RGB (255, 165, 0) and RGB (255, 105, 180) to HLS.**
 - Compute the **Hue**, **Lightness**, and **Saturation** for both colors.